

A comparative analysis of Von-Heijne, SVM and Perceptron Models for the detection of Signal Peptides.

Abstract

Motivation: This is the motivation

Results: This are the results.

Introduction

For a protein to enter the secretory pathway, in both eukaryotic and prokaryotic cells, it must be endowed with a specific target signal. Often, this signal takes the shape of a short sequences located at the N-terminus of proteins ref. The signal peptide has a distinct three-domain structure, depicted in Figure 1, with a positively charged N-terminal region (n-region), a central hydrophobic region (h-region) and a more polar C-terminal region (c-region) containing the cleavage site ref. Given their importance in many aspects of cell biology, the accurate detection of SPs is a crucial task in bioinformatics, which has been tackled before with a variety of approaches, including machine learning (SVMs ref, ref and Bayesian networks ref) and deep learning ref.

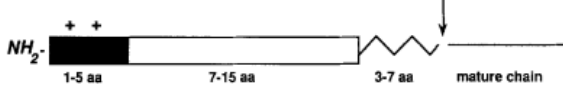


Figure 1: Signal peptide structure

With this work, we propose a comparative analysis of three different approaches for the detection of SPs, namely the Von-Heijne model, a Support Vector Machine (SVM) and a Multi-layer Perceptron (MLP).

Materials and Methods

Data Collection

Reviewed non-fragment protein sequences from the eukaryota taxa were queried from UniprotKB, and collected into 2 distinct sets, positive and negative. For Positive sequences, we selected sequences having experimental evidence for the presence of a signal peptide (ft_signal_exp:*) For the negative set, proteins not annotated with a signal peptide and endowed with experimental evidence for subcellular localization in non-secretory compartments, namely nucleus (cc_scl_term_exp:SL-0191), peroxisome (cc_scl_term_exp:SL-0204), cell membrane /

plasma membrane (cc_scl_term_exp:SL-0039), cytosol (cc_scl_term_exp:SL-0091), plastid (cc_scl_term_exp:SL-0209) or mitochondrion (cc_scl_term_exp:SL-0173). For sequences in the negative set, we gathered information about the presence of a transmembrane helix, whose hydrophobicity profile mimicks the one of a signal peptide.

Further selection on the positive entries was carried out in order to select proteins which had informaton about the cleavage site and a Signal Peptide with a length of at least 14bp.

Clustering

Sequences were clustered using mmseqs2 ref, using parameters --min-seq-id 0.3, cluster-mode 1 -cov-mode 0 -c 0.4, in order to cluster sequences with a similarity of at least 30% and an alignment spanning at least 40% of the total sequence length. The number of sequences in the datasets is reported in Table 1.

Dataset	Before Clustering	After Clustering
Positive	2942	1093
Negative (total)	20806	9028
Negative (w/o TM)	18301	8113
Negative (with TM)	2505	915

Table 1: Dataset sizes before and after clustering

Von-Heijne Model

The full dataset was split into a training and a test set with a ratio of 0.8. A Position-Specific Weight Matrix (PSWM) was then generated using 16bp windows (13bp upstream to 2bp downstream) surrounding the cleavage site. Amino acid frequencies were normalized using background amino acid frequencies from SwissProt. After initializing the matrix with pseudocounts of 1, the weights are computed as follows:

$$w(a, i) = \log_2 \left(\frac{f(a, i)}{f(a)} \right)$$

Where $f(a, i)$ is the frequency of amino acid a at position i in the training set, and $f(a)$ is the background frequency of amino acid a in SwissProt. The score for a given sequence is then computed as:

$$S = \max_{j=1}^{90-16+1} \sum_{i=1}^{16} w(a_{j+i-1}, i)$$

scores are computed over the first 90bp of the given sequence, using a sliding window 16bp long. Then, a sequence was labeled as positively endowed with a SP if the Score S is greater than a threshold t .

$$\text{label} = \begin{cases} \text{positive} & \text{if } S > t \\ \text{negative} & \text{if } S \leq t \end{cases}$$

The optimal threshold for the dataset was computed using 5-fold cross validation.

Feature Extraction

To build a model that can deal with sequences, they must be encoded into numerical representation. To do so, we extracted numerical feature values from the sequence. We used 2 kinds of features:

- Aminoacid composition up to a cutoff k (n_term_composition) and from k to an hard-coded limit (set to 90) (c_term_composition)
- Protein Scales computed on the first 90 aminoacids with a window size of 5 (Figure 2). An Hydrophobicity scale (cit), alpha-helix tendency scale (cit), transmembrane tendency (cit), bulkiness (cit), and polarity (cit) were used.

For this experiment, k was chosen to be the average position of the cut site, roughly 22 aminoacids from the N-Terminus.

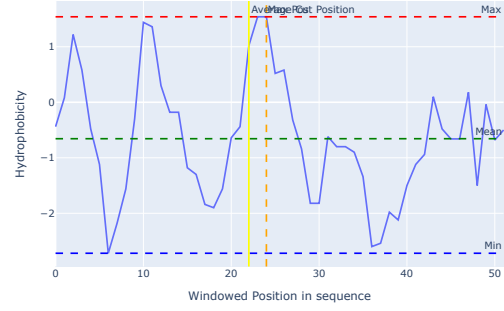


Figure 2: Confusion matrix for the Von-Heijne model

Support Vector Machine

We used a grid search to find optimal hyperparameters, testing both RBF and linear kernels. For the RBF kernel, we evaluated C values of 0.1, 1, 10, and 100, combined with gamma values of “scale”, 0.001, 0.01, 0.1, and 1. For the linear kernel, we tested C values of 0.1, 1, 10, and 100.

Results

SP Motifs

From all SP-Endowed sequences, a motif logo was generated using the context around the cleavage site (13 bp upstream to 2bp downstream) Figure 3.

For the Von-Heijne model the computed optimal threshold was 9.25, confusion matrix for the test set is reported in Figure 4.

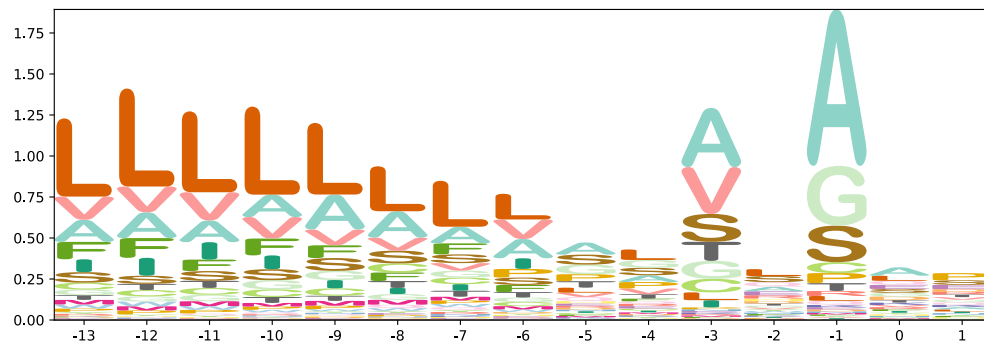


Figure 3: Motif logo for the context (-13bp to 2bp) surrounding the cleavage site

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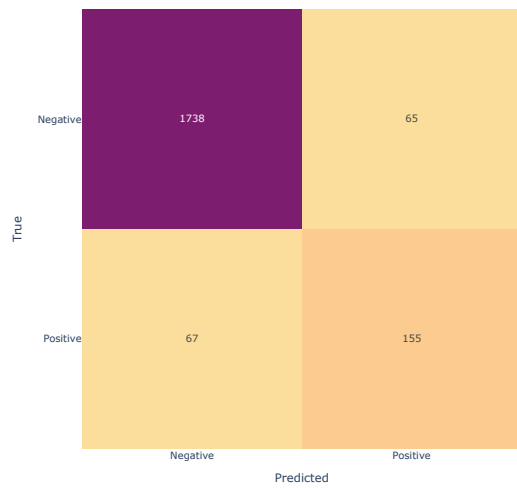


Figure 4: Confusion matrix for the Von-Heijne model